

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)

2025-26, Term-III

Project Problem Statement Submission Form

Sec No- 12	Team No- 7	Project Title- Flight Comparison Tool
Team Lead Roll Number	2520030601	
Team Lead Name	Billakanti Siri	
Team Member Roll Number	2520090010	
Team Member Name	Varnika Komali	

1. Abstract

This project proposes a Flight Comparison Tool that provides a comprehensive and user-centric platform to evaluate flights across multiple dimensions.

The system integrates real-time and historical data to enable users to compare flights based on price trends, airline performance, baggage allowances, and route efficiency. A dynamic date price calendar allows users to identify the most cost-effective travel dates, while an airline comparison module highlights differences in services and reliability metrics.

The application features an interactive filter and sorting system to refine results based on user preferences such as cost, duration, and punctuality. Additional modules include a baggage policy comparison grid, on-time performance statistics, and route visualisation.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
UI Design & Responsive Layout	2520030601 2520090010	
Flight listing Interface	2520030601 2520090010	
Filter & Sort System	2520030601	
Baggage Policy Grid	2520030601	
Airline Comparison Module	2520030601	
Date-Based Price View	2520090010	
Routing and Navigation	2520090010	
Flight Data Integration	2520090010	

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

Student 1 – Roll No: 2520030601

- Responsible for designing the frontend interface of the Flight Comparison Tool.
- Developed a responsive and user-friendly UI, ensuring smooth usage across desktop and mobile devices.
- Implemented the flight listing interface, displaying flight details such as airline, price, and duration in a structured format.
- Designed and developed the filter and sort system, allowing users to refine results based on:
 - Price range
 - Flight duration
 - Preferred airline
- Created the baggage policy grid, presenting airline-specific baggage rules in a clear and comparable manner.
- Contributed to the airline comparison module by organising and structuring the visual layout for easy comparison.
- Focused on improving usability, layout design, and interactive elements of the application.

Student 2 – Roll No: 2520090010

- Responsible for data handling and core application logic of the system.
- Implemented the flight data integration module, including:
 - Fetching flight data (via API or mock dataset)
 - Managing and structuring data for use in the application
- Developed the date-based price view, enabling users to:
 - Compare flight prices across different dates
 - Identify cost-effective travel options
- Handled routing and navigation, ensuring smooth transitions between pages such as:
 - Search results page
 - Flight comparison page
- Ensured proper integration between frontend and data modules, maintaining consistency in displayed information.
- Collaborated on UI design and flight listing interface for seamless functionality.
- Focused on application flow, data processing, and backend logic implementation.

4. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Frameworks, Libraries, Platforms, and Tools Used

- React.js – For building the user interface
- JavaScript (ES6+) – For application logic
- CSS / Bootstrap / Tailwind CSS – For styling and responsive design
- React Router – For navigation between pages
- Axios / Fetch API – For data handling (API or JSON data)
- Node.js & Express.js (optional) – For backend development
- MongoDB / JSON (optional) – For data storage
- VS Code – Development environment
- Git & GitHub – Version control

Guide Name and Signature (remarks if any) _____